Daily Algebra Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Geometric series identity · Polynomial remainder shortcut · Parity and mod arguments · AM–GM under substitution.

Section A Rapid Recognition

1. Evaluate without expanding: 201×199.

Ans: 39,999

2. Factorise: $x^2 - x - 42$.

Ans: $(x-7)(x+6)$

3. Simplify $1 + x + x^2 + x^3 + x^4$ at $x = -1$.

Ans: 1

4. Given $p + q = 6$, $pq = 7$, find $p^3 + q^3$.

Ans: 90

5. Remainder when $f(x) = x^3 + 2x - 3$ is divided by $(x - 1)$.

Ans: 0

6. Factorise completely: $2x^3 - 18x$.

Ans: $2x(x-3)(x+3)$

7. Simplify: $\dfrac{3^{n+1} - 3^{n-1}}{3^n}$.

Ans: $\frac{8}{3}$

8. Given $a - b = 3$ and $a^2 + b^2 = 29$, find ab.

Ans: $ab = 10$

9. Factorise $x^4 - 1$ completely over $\mathbb{R}$.

Ans: $(x^2 + 1)(x + 1)(x - 1)$

10. Simplify: $\dfrac{n!}{(n-2)!}$.

Ans: $n(n - 1)$

Section B Manipulation Drills

1. Simplify $\dfrac{x^5 - 1}{x - 1}$ using the geometric series formula.

Ans: $x^4 + x^3 + x^2 + x + 1$

2. Find the remainders when $g(x) = x^{10} - 3x^5 + 2$ is divided by $(x - 1)$ and $(x + 1)$.

Ans: Remainder at $(x - 1)$: $g(1) = 1 - 3 + 2 = 0$. Remainder at $(x + 1)$: $g(-1) = 1 + 3 + 2 = 6$.

3. Simplify: $\dfrac{ab}{a - b} - \dfrac{a^2 b}{a^2 - b^2}$.

Ans: $\dfrac{ab^2}{a^2 - b^2}$

4. Factorise $x^6 - y^6$ completely over $\mathbb{Z}$.

Ans: $(x-y)(x+y)(x^2+xy+y^2)(x^2-xy+y^2)$

5. Roots of $3x^2 + 5x - 2 = 0$ are α, β. Find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Ans: $-\frac{5}{2}$

6. Simplify: $\left(1 - \frac{1}{x^2}\right)(x^2 + x)$.

Ans: $\frac{x}{(x-1)(x+1)^2}$

7. Given $a+b+c=5$, $ab+bc+ca=6$, find $a^2+b^2+c^2$.

Ans: 13

8. Rearrange for y: $\dfrac{2y+1}{y-3}=x$.

Ans: $y=\dfrac{3x+1}{x-2}$

9. Show $4n^2-1=(2n-1)(2n+1)$; hence prove $4n^2-1$ is never prime for $n\geq 2$.

Ans: For $n\geq 2$: $2n-1\geq 3$ and $2n+1\geq 5$, both greater than 1, so $4n^2-1$ is a product of two integers each >1. Hence composite.

10. With $\alpha + \beta = 4$, $\alpha\beta = 1$, find the quadratic with integer coefficients whose roots are $\alpha - \frac{1}{\alpha}$ and $\beta - \frac{1}{\beta}$.

Ans: $x^2 - 12 = 0$

Section C Substitution & Structure

1. Solve: $4^x - 5 \cdot 2^x + 4 = 0$.

Ans: $x = 0$ and $x = 2$

2. Express $x^3 + \frac{1}{x^3}$ in terms of $t = x + \frac{1}{x}$.

Ans: $t^3 - 3t$

3. Find the minimum of $3x + \frac{27}{x}$ for $x > 0$.

Ans: Minimum is 18 at $x = 3$.

4. Solve for real x: $\sqrt{2x-1} + \sqrt{x-1} = 2$.

Ans: $x = 12 - 4\sqrt{7}$ (one real solution)

5. Find the value of k such that $(x+1)(x+3)(x+5)(x+7) + k$ is a perfect square.

Ans: $k = 16$

6. Prove $(a^2+b^2)(c^2+d^2) \geq (ac+bd)^2$. State the equality condition.

Ans: Equality iff $ad = bc$.

7. With $u = x+y$, $v = x-y$, express x^4+y^4 in terms of u and v.

Ans: $\frac{u^4+v^4+6u^2v^2}{8}$

8. Solve: $x^4 - 7x^2 + 12 = 0$.

Ans: $x = \pm 2,\ \pm\sqrt{3}$

Section D Challenge (TMUA / SMC difficulty)

1. Prove $n^2 \equiv 0$ or $1 \pmod 4$ for all integers n; hence $n^2 + 2$ is never divisible by 4.

Ans: Proved below.

2. Find all real solutions to $x^4 - 4x^3 + 4x^2 - 1 = 0$.

Ans: $x = 1$ (double) and $x = 1 \pm \sqrt{2}$

3. Sequence: $a_1 = 1$, $a_{n+1} = \frac{a_n}{a_n+2}$. Find a closed form for a_n and a_{100}.

Ans: $a_n = \frac{1}{2n-1}$; $a_{100} = \frac{1}{2^{100}-1}$

4. Evaluate $\prod_{k=2}^{n}\left(1 - \frac{1}{k^2}\right)$ in closed form. Find its limit as $n \to \infty$.

Ans: $\frac{n+1}{2n}$; $\text{limit} = \frac{1}{2}$

5. $p(x) = x^4 + ax^3 + bx^2 + cx + d$ with $p(1) = 10$, $p(2) = 20$, $p(3) = 30$, $p(4) = 40$. Find $p(12) - 12p(0)$.

Ans: 7752